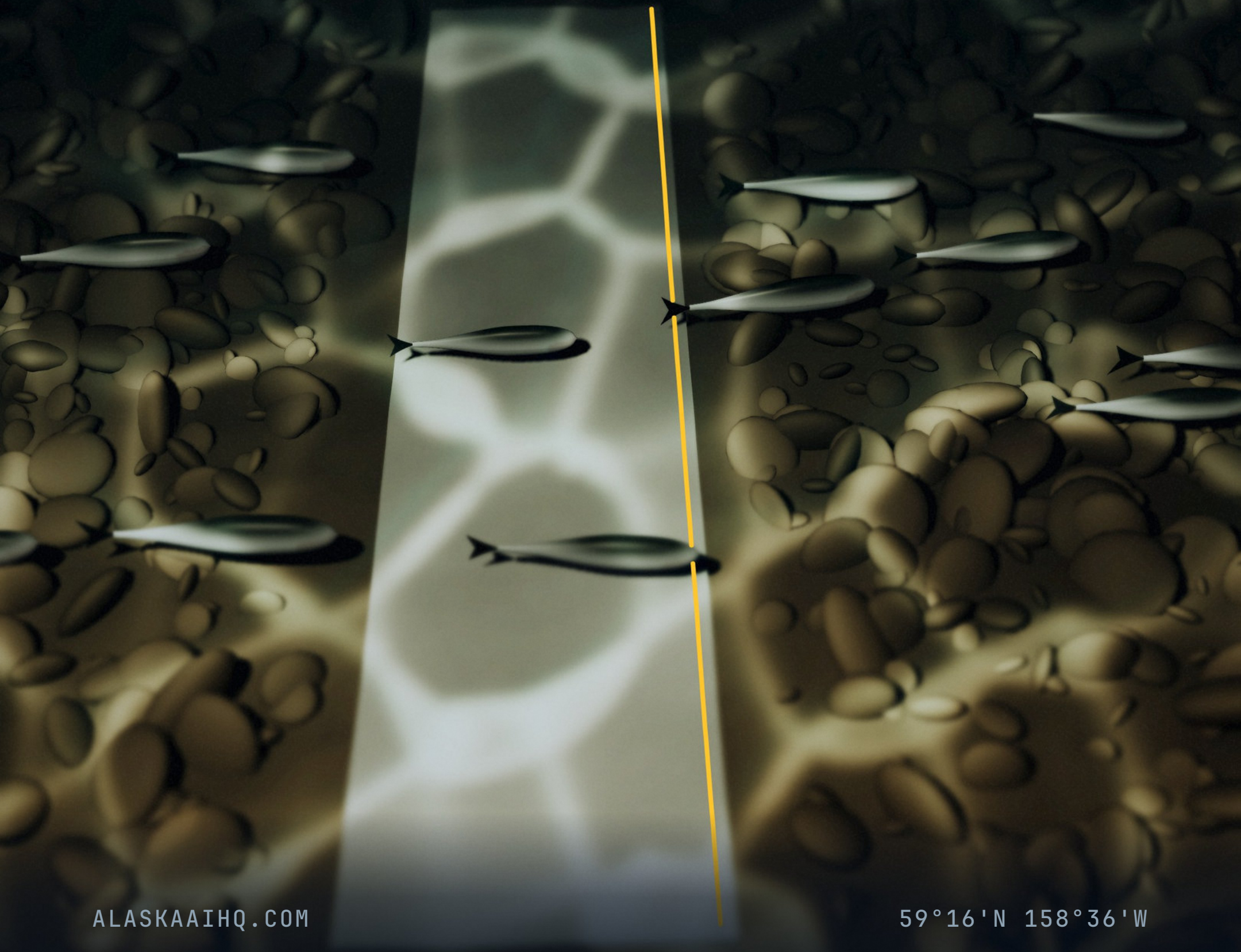


BRISTOL BAY • WHAT THE AI DRONE WILL BE GRADED ON

Togiak missed by 702 fish. Its tower counted 10 minutes an hour.



HOW A TOWER COUNTS

Ten minutes, times six.

Bristol Bay's towers count each bank for 10 minutes of every hour, then expand. Crews work around the clock with hand clickers. At six to one, the Togiak's 119,298 rests on roughly 19,883 fish someone saw.

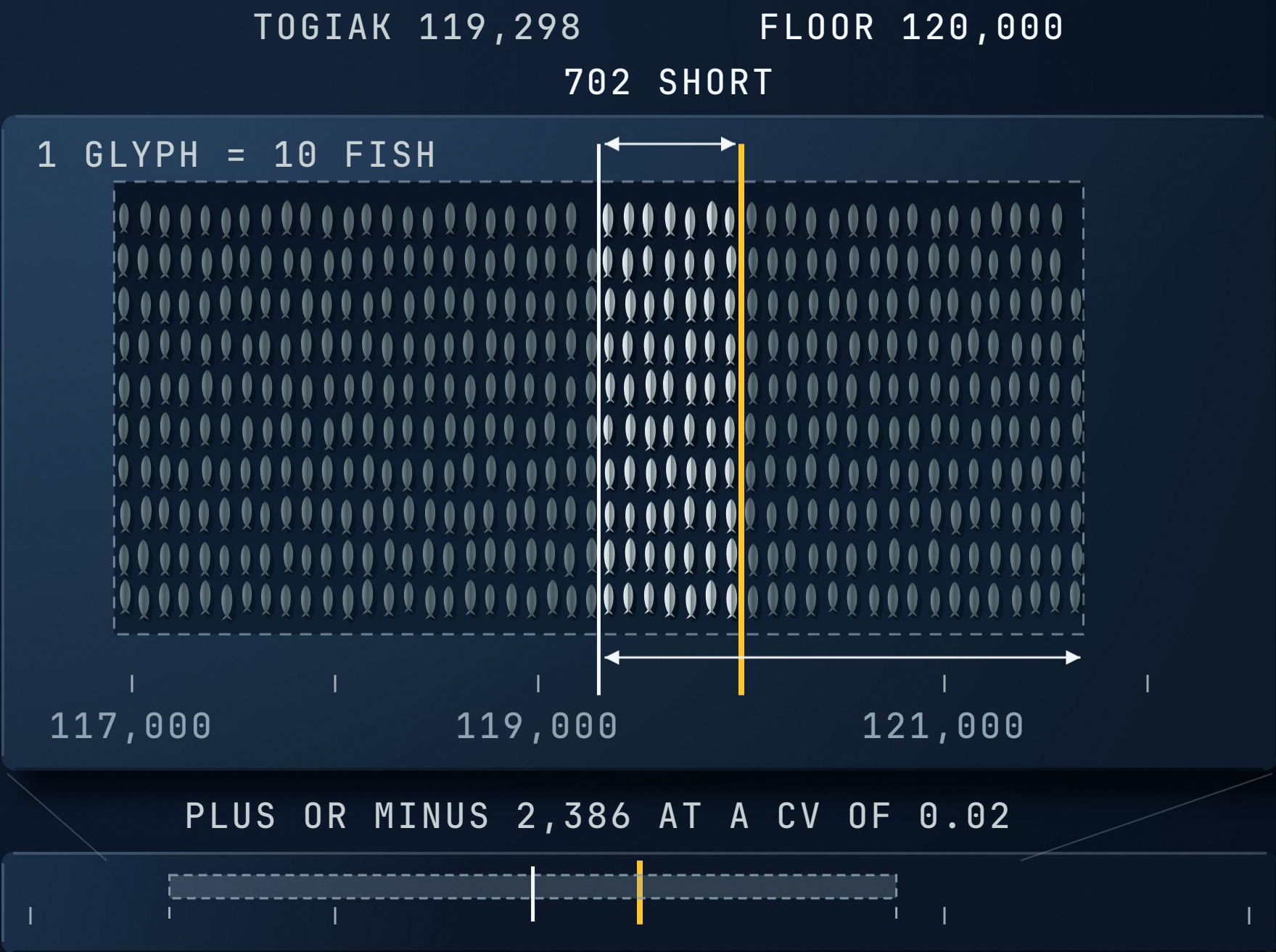


1 OF 6 TEN-MINUTE WINDOWS

THE TOWER'S OWN NUMBER

The error bar runs three times the miss.

ADF&G put the average sampling CV across nine Bristol Bay tower projects at 0.02 in 2004 and 2005. On the Togiak's 119,298 that is about 2,400 fish either way. It assumes no miscounts inside the ten minutes.



WAS IT EVER CHECKED

The rule was tested on 12 to 80 hours.

In 1965 and 1966, crews counted whole hours and compared them with the first 10 minutes, expanded. Errors ran generally under 10 percent, with no significant bias.

ONE HOUR



12 TO 80 HOURS COUNTED IN FULL ONE MONTH, 720 HOURS

2026 SOCKEYE ESCAPEMENT, PRELIMINARY

Eight towers counted

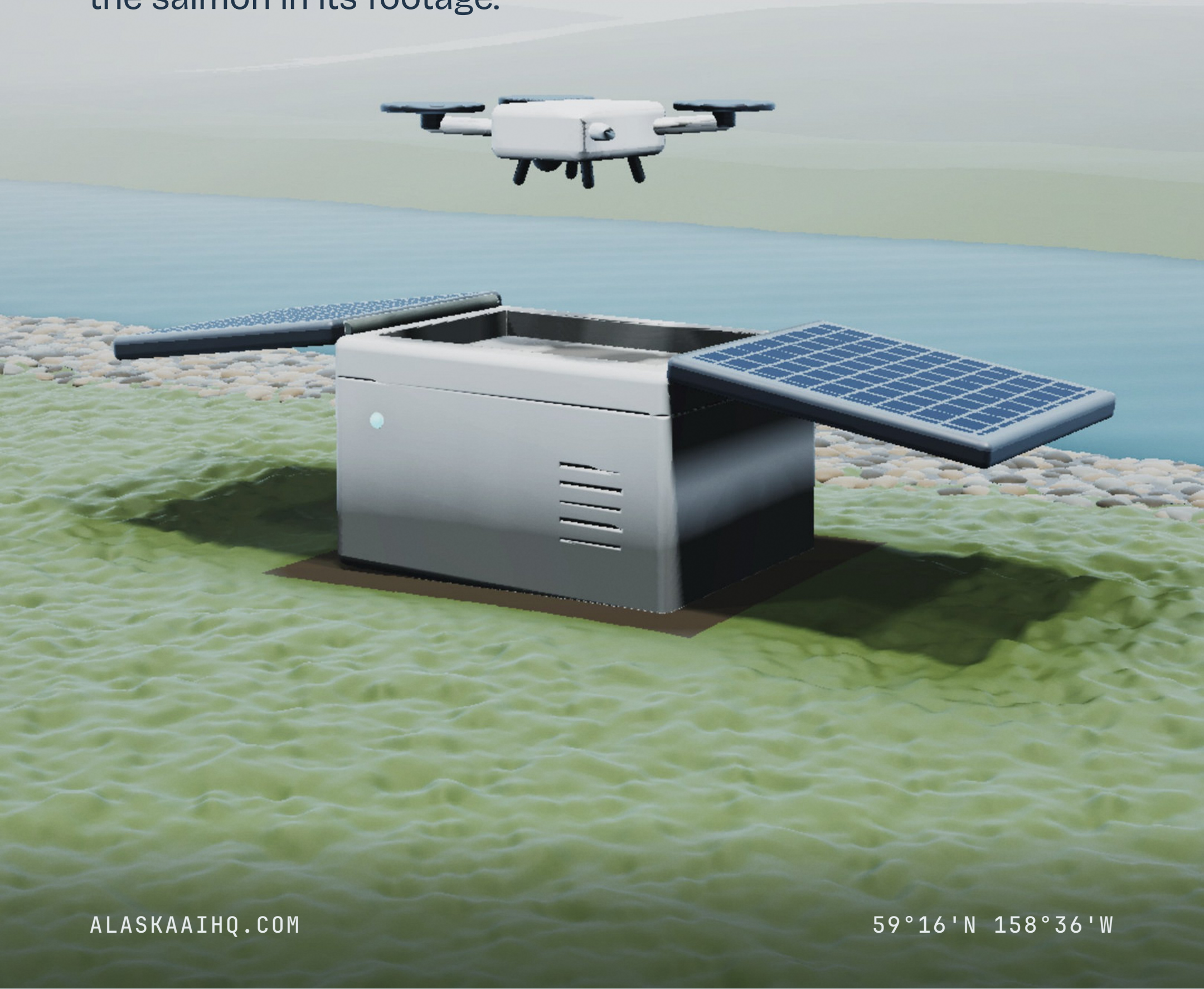
12.3 million of 15.6 million.



WOOD RIVER • 8:30 A.M.

A box opens. A drone takes off.

A computer vision model is learning to pick out the salmon in its footage.



WHAT THE MODEL LEARNS FROM

A key press for every fish.

Technicians press a key each time a fish crosses a line drawn on the footage. The first phase logged 256 flights and 17,552 images, stitched into 117 orthomosaics. The project lead says it spots fish 12 to 18 hours before the tower.

A UW, BBRSDA AND BBEDC PILOT. FISH AND GAME NOT DIRECTLY INVOLVED AS OF JULY



THE TEST IS NEXT

Matching the tower can't be the bar.

The pilot plans to test its model against Fish and Game tower counts next. Tower sampling error averaged about 2 percent in 2004 and 2005. The team's goal is a count with a stated confidence. Write the margin first.

PASS IF WITHIN	
SET BY	

Before the Wood River test runs, who should set the passing margin?